

1.Install MariaDB on your Linux system and verify the installation by running the command to check the MariaDB server version.

Ans-

Step 1: Update packages: `sudo dnf update -y`

Step 2: Install MariaDB Server: `sudo dnf install mariadb-server`

-y Step 3: Start MariaDB Service: `sudo systemctl start mariadb`

Step 4: Enable MariaDB at Boot: `sudo systemctl enable`

`mariadb` Step 5: Check Service Status: `sudo systemctl`

`status mariadb` Step 6: Verify MariaDB Version: `mariadb --`

`version`

```

root@server1:~ — sudo dnf install mariadb-server -y

[root@server1 ~]# sudo dnf update -y
Last metadata expiration check: 0:27:47 ago on Tue 04 Aug 2026 01:52:04 AM EDT.
Dependencies resolved.
=====
Package           Arch      Version      Repository      Size
=====
Upgrading:
docker-compose-plugin x86_64    5.4.0-1.el9   docker-ce-stable 12 M

Transaction Summary
=====
Upgrade 1 Package

Total download size: 12 M
Downloading Packages:
docker-compose-plugin-5.4.0-1.el9.x86_64.rpm 328 kB/s | 12 MB 00:35
-----
Total                                         328 kB/s | 12 MB 00:36
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Upgrading      : docker-compose-plugin-5.4.0-1.el9.x86_64 1/2
  Running scriptlet: docker-compose-plugin-5.4.0-1.el9.x86_64 1/2
  Running scriptlet: docker-compose-plugin-5.4.0-1.el9.x86_64 2/2

```

```
[root@server1 ~]# sudo systemctl start mariadb
[root@server1 ~]# sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@server1 ~]# sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-08-04 02:23:00 EDT; 16s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
  Main PID: 5691 (mariabdd)
    Status: "Taking your SQL requests now..."
     Tasks: 11 (limit: 41856)
    Memory: 73.0M (peak: 105.3M)
       CPU: 1.452s
    CGroup: /system.slice/mariadb.service
            └─5691 /usr/libexec/mariabdd --basedir=/usr

Aug 04 02:23:00 server1.tops.com mariadb-prepare-db-dir[5648]: The second is mysql@localhost, it has>
Aug 04 02:23:00 server1.tops.com mariadb-prepare-db-dir[5648]: you need to be the system 'mysql' use>
Aug 04 02:23:00 server1.tops.com mariadb-prepare-db-dir[5648]: After connecting you can set the pass>
Aug 04 02:23:00 server1.tops.com mariadb-prepare-db-dir[5648]: able to connect as any of these users>
Aug 04 02:23:00 server1.tops.com mariadb-prepare-db-dir[5648]: See the MariaDB Knowledgebase at http>
```

```
Aug 04 02:23:00 server1.tops.com systemd[1]: Started MariaDB 10.5 database server.
[root@server1 ~]# mariadb --version
mariadb Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper
[root@server1 ~]#
```

2. Create a new MariaDB database named foodie_app and a table restaurants with columns id (INT, primary key), name (VARCHAR), cuisine (VARCHAR), and rating (FLOAT). Insert 3 sample restaurant records.

Ans-

Step 1: Login to MariaDB: sudo

mariadb You should see: MariaDB

[(none)]>

Step 2: Create Database

CREATE DATABASE foodie_app;

Verify: SHOW DATABASES;

Step 3: Select Database: USE foodie_app;

Step 4: Create Table

CREATE TABLE restaurants

(id INT PRIMARY KEY,

name VARCHAR(100),

cuisine VARCHAR(50),

rating FLOAT

);

Verify: SHOW TABLES;

Step 5: View Table Structure: DESCRIBE

restaurants; Step 6: Insert Sample Records

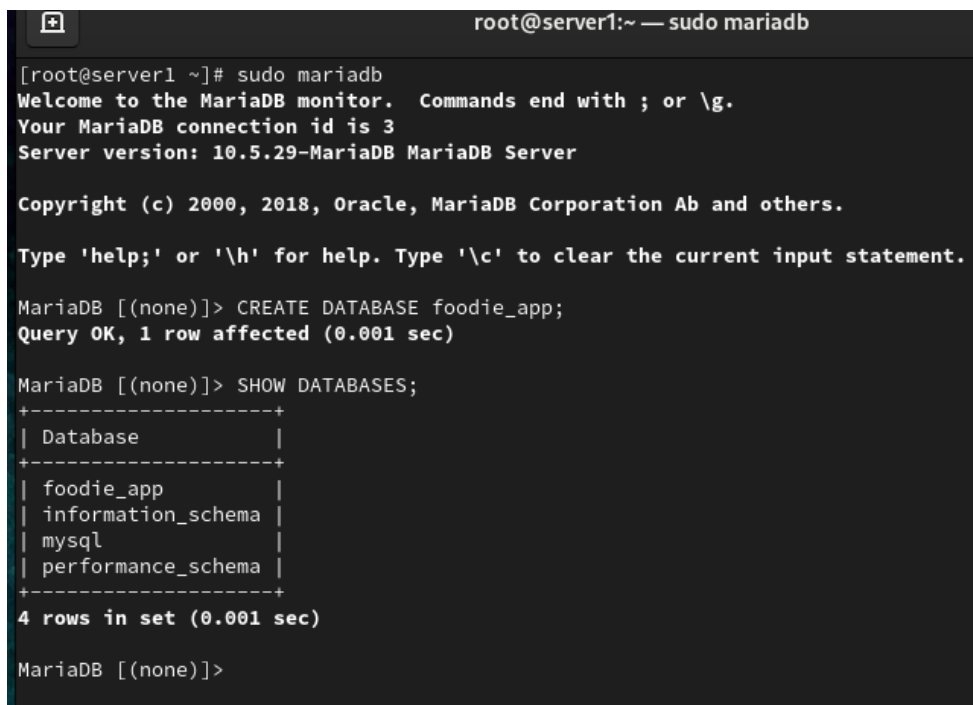
INSERT INTO restaurants VALUES

(1,'Pizza Palace','Italian',4.8),

(2,'Spice Villa','Indian',4.2),

(3,'Dragon House','Chinese',3.9);

Step 7: Verify Records: SELECT * FROM restaurants;



```
root@server1:~ — sudo mariadb

[root@server1 ~]# sudo mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE foodie_app;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database                |
+-----+
| foodie_app               |
| information_schema       |
| mysql                    |
| performance_schema       |
+-----+
4 rows in set (0.001 sec)

MariaDB [(none)]>
```

```

MariaDB [(none)]> USE foodie_app;
Database changed
MariaDB [foodie_app]> CREATE TABLE restaurants (
  ->     id INT PRIMARY KEY,
  ->     name VARCHAR(100),
  ->     cuisine VARCHAR(50),
  ->     rating FLOAT
  -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [foodie_app]> SHOW TABLES;
+-----+
| Tables_in_foodie_app |
+-----+
| restaurants           |
+-----+
1 row in set (0.001 sec)

```

```

4 rows in set (0.002 sec)

MariaDB [foodie_app]> INSERT INTO restaurants VALUES
  -> (1,'Pizza Palace','Italian',4.8),
  -> (2,'Spice Villa','Indian',4.2),
  -> (3,'Dragon House','Chinese',3.9);
Query OK, 3 rows affected (0.008 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [foodie_app]> SELECT * FROM restaurants;
+-----+-----+-----+-----+
| id | name       | cuisine | rating |
+-----+-----+-----+-----+
| 1 | Pizza Palace | Italian | 4.8    |
| 2 | Spice Villa  | Indian  | 4.2    |
| 3 | Dragon House | Chinese | 3.9    |
+-----+-----+-----+-----+
3 rows in set (0.001 sec)

MariaDB [foodie_app]>

```

3. Write and execute a SQL query in MariaDB to find all restaurants in the foodie_app database with a rating greater than 4.0 and display their name and cuisine.

Ans-

Find Restaurants Rating Greater Than 4 Run

SELECT name,cuisine

FROM restaurants

WHERE rating > 4.0;

```
MariaDB [foodie_app]> SELECT name,cuisine
-> FROM restaurants
-> WHERE rating > 4.0;
+-----+-----+
| name      | cuisine |
+-----+-----+
| Pizza Hut | Italian |
| Spice Villa | Indian |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [foodie_app]>
```

4. Grant **SELECT** and **INSERT** permissions on the **foodie_app** database to a new MariaDB user named **zomato_user** with password **zomato123**, then test the permissions by connecting as that user and inserting a new restaurant.

Ans-

Step 1: Create User

```
CREATE USER 'zomato_user'@'localhost'
IDENTIFIED BY 'zomato123';
```

Step 2: Grant Permissions

```
GRANT SELECT, INSERT
```

```
ON foodie_app.*
```

```
TO 'zomato_user'@'localhost';
```

Step 3: Reload Privileges: **FLUSH PRIVILEGES;**

Step 4: Verify Grants: **SHOW GRANTS FOR**

'zomato_user'@'localhost'; Step 5: Exit MariaDB: **EXIT;**

Step 6: Login as New User: **mariadb -u**

zomato_user -p Step 7: Select Database: **USE**

foodie_app;

Step 8: Insert New Restaurant

```
INSERT INTO restaurants
```

```
VALUES (4,'Burger Hub','Fast
```

```
Food',4.4);
```

 Step 9: Verify Insert

```
SELECT * FROM restaurants;
```

```

MariaDB [(none)]> CREATE USER 'zomato_user'@'localhost'
-> IDENTIFIED BY 'zomato123';
Query OK, 0 rows affected (0.117 sec)

MariaDB [(none)]> GRANT SELECT, INSERT
-> ON foodie_app.*
-> TO 'zomato_user'@'localhost';
Query OK, 0 rows affected (0.046 sec)

MariaDB [(none)]>

```

```

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.061 sec)

MariaDB [(none)]> SHOW GRANTS FOR 'zomato_user'@'localhost';
+-----+
| Grants for zomato_user@localhost |
+-----+
| GRANT USAGE ON *.* TO `zomato_user`@`localhost` IDENTIFIED BY PASSWORD '*68150B860C64CC67FDBFE80D9DE912BACE50105C' |
| GRANT SELECT, INSERT ON `foodie_app`.* TO `zomato_user`@`localhost` |
+-----+
2 rows in set (0.007 sec)

```

5. Use ChatGPT or Copilot to generate a SQL query that updates the rating of all restaurants with cuisine 'Italian' to 4.5 in your MariaDB table, then run the query and verify the update. **Hint:** Paste your prompt and the AI-generated query in your submission along with the result.

Ans-

Update Italian Restaurants Rating Prompt Used Write a SQL query to update the rating of all restaurants with cuisine 'Italian' to 4.5.

AI Generated

Query UPDATE

restaurants SET

rating = 4.5

WHERE cuisine='Italian';

Verify : SELECT * FROM

restaurants;

```
MariaDB [foodie_app]> SELECT * FROM restaurants;
```

id	name	cuisine	rating
1	Pizza Hut	Italian	4.5
2	Dominos	Italian	4.5
3	Spice Villa	Indian	4.8
4	Burger King	Fast Food	4.1

```
4 rows in set (0.001 sec)
```

```
MariaDB [foodie_app]> 
```